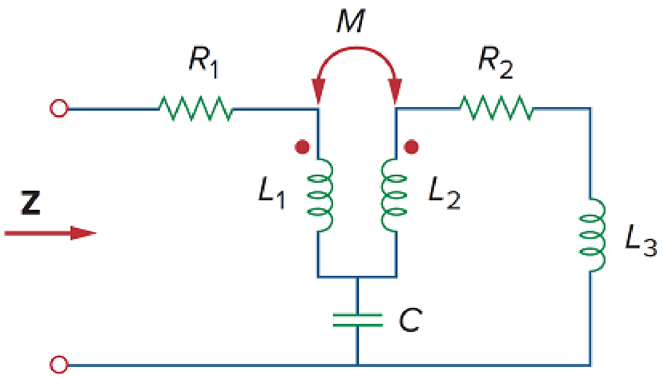
Problem

Using Fig. 13.103, design a problem to help other students better understand how to find the input impedance of circuits with transformers.

Figure 13.103



Step-by-step solution

Step 1 of 4

Refer to circuit diagram in Figure 13.103 in the textbook.

Draw the frequency domain circuit.

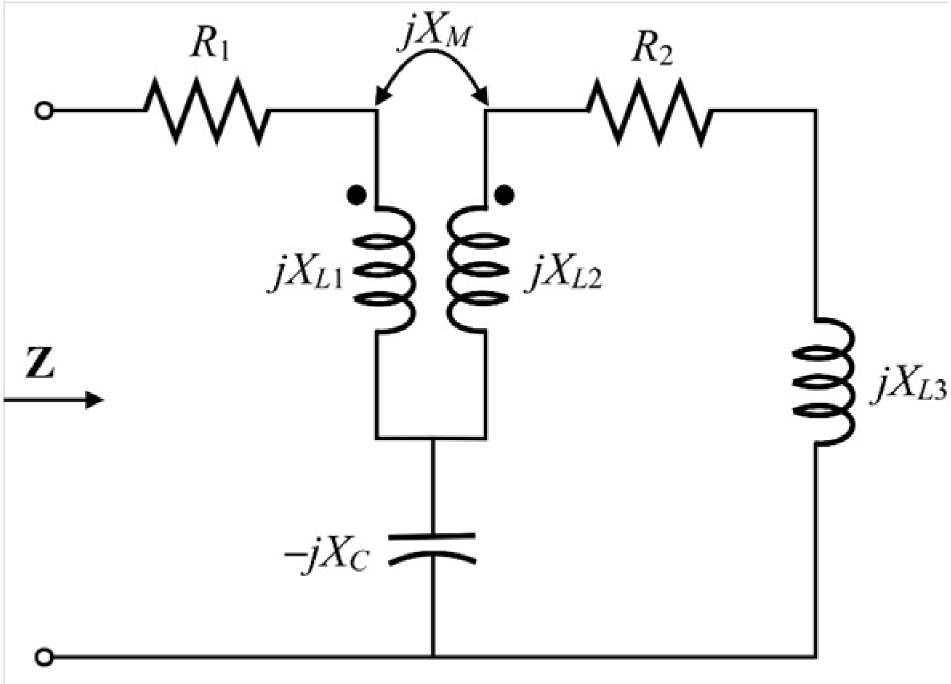


Figure 1

Step 2 of 4

Now, convert linear transformer into an equivalent T circuit. The circuit diagrams of linear transformer and equivalent T network is shown in Figure 2.

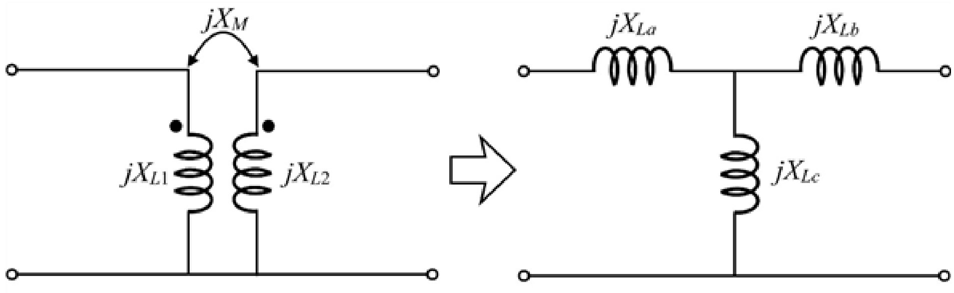


Figure 2

Step 3 of 4

Calculate the value of impedance jX_{La}.

$$jX_{La} = jX_{L1} - jX_M$$

Calculate the value of impedance jX_{Lb}.

$$jX_{Lb} = jX_{L2} - jX_M$$

Calculate the value of impedance jX_{Lc}.

$$jX_{Lc} = jX_M$$

The modified circuit diagram is shown in Figure 3.

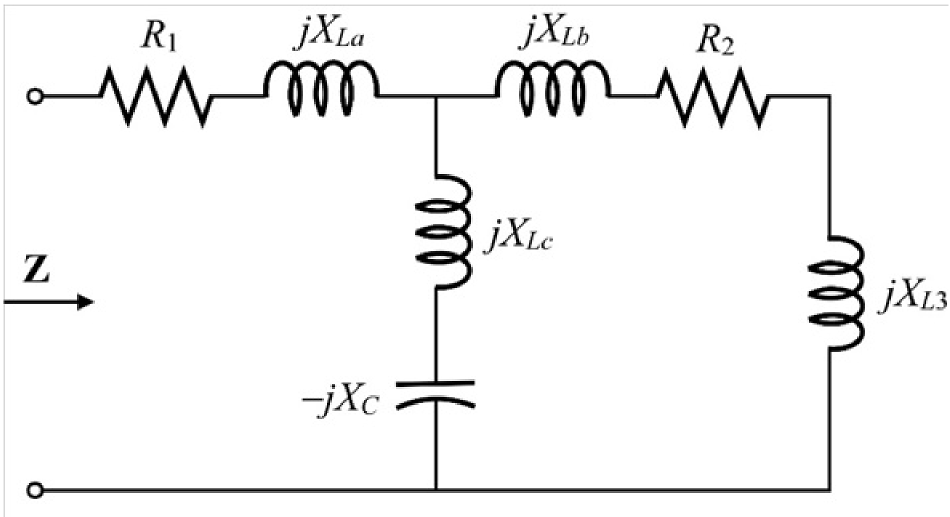


Figure 3

Step 4 of 4

Calculate the input impedance of the circuit.

$$\mathbf{Z} = R_1 + jX_{La} + \left(jX_{Lc} - jX_C\right) \| \left(jX_{Lb} + R_2 + jX_{L3}\right)$$

Substitute $jX_{L1} - jX_M$ for jX_{La}, $jX_{L2} - jX_M$ for jX_{Lb} and jX_M for jX_{Lc}.

$$\begin{aligned}
\mathbf{Z} &= R_1 + jX_{L1} - jX_M + \left(jX_M - jX_C\right) \| \left(jX_{L2} - jX_M + R_2 + jX_{L3}\right) \\
&= R_1 + jX_{L1} - jX_M + \frac{\left(jX_M - jX_C\right)\left(jX_{L2} - jX_M + R_2 + jX_{L3}\right)}{jX_M - jX_C + jX_{L2} - jX_M + R_2 + jX_{L3}} \\
&= R_1 + jX_{L1} - jX_M + \frac{\begin{pmatrix} -X_{L2}X_M + X_M^2 + jR_2X_M - X_{L3}X_M + \\ X_{L2}X_C - X_MX_C - jR_2X_C + X_{L3}X_C \end{pmatrix}}{-jX_C + jX_{L2} + R_2 + jX_{L3}} \\
&= \frac{\left(R_1 + jX_{L1} - jX_M\right)\begin{pmatrix} -jX_C + jX_{L2} + \\ R_2 + jX_{L3} \end{pmatrix} + \begin{pmatrix} -X_{L2}X_M + X_M^2 + jR_2X_M - X_{L3}X_M + \\ X_{L2}X_C - X_MX_C - jR_2X_C + X_{L3}X_C \end{pmatrix}}{-jX_C + jX_{L2} + R_2 + jX_{L3}} \\
&= \frac{\begin{pmatrix} -jR_1X_C + jR_1X_{L2} + R_1R_2 + jR_1X_{L3} + \\ X_{L1}X_C - X_{L1}X_{L2} + jR_2X_{L1} - X_{L1}X_{L3} \\ -X_MX_C + X_MX_{L2} - jR_2X_M + X_MX_{L3} \end{pmatrix} + \begin{pmatrix} -X_{L2}X_M + X_M^2 + jR_2X_M - X_{L3}X_M + \\ X_{L2}X_C - X_MX_C - jR_2X_C + X_{L3}X_C \end{pmatrix}}{-jX_C + jX_{L2} + R_2 + jX_{L3}} \\
&= \frac{\begin{pmatrix} jR_1\left(X_{L2} + X_{L3} - X_C\right) + jR_2\left(X_{L1} - X_C\right) + R_1R_2 + \\ X_{L1}\left(X_C - X_{L2} - X_{L3}\right) + X_M\left(X_M - 2X_C\right) + X_C\left(X_{L2} + X_{L3}\right) \end{pmatrix}}{-jX_C + jX_{L2} + R_2 + jX_{L3}}
\end{aligned}$$

Therefore, the input impedance, $\mathbf{Z}$ of the circuit is,

$$\boxed{\frac{\begin{pmatrix} jR_1\left(X_{L2} + X_{L3} - X_C\right) + jR_2\left(X_{L1} - X_C\right) + R_1R_2 + \\ X_{L1}\left(X_C - X_{L2} - X_{L3}\right) + X_M\left(X_M - 2X_C\right) + X_C\left(X_{L2} + X_{L3}\right) \end{pmatrix}}{-jX_C + jX_{L2} + R_2 + jX_{L3}}}.$$